

Asset Pricing Theory

MFE5340: AI in FE: Quantitative Investment

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1 Goals

- Goals
- Asset Returns
- Mean-Variance Analysis
- Stochastic Discount Factor
- Multi-Factor Models
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- Factor Investing: An Overview



Set the Tone

1 Goals

- The content covered in this lecture is by no means a full treatment of the relevant concepts in asset pricing. A full treatment is beyond the scope of this course.
- My goal is to provide you with enough background knowledge and intuition to conduct investment practice in industry, and to understand the latest empirical asset pricing papers.
- I will introduce the relevant concepts and explain the equivalence between the SDF, mean-variance frontier, and factor models.
- For a comprehensive and systematic introduction to asset pricing, I strongly recommend reading Cochrane (2005) carefully.

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2 Asset Returns

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- Let P_{it} denote the price of some asset i (e.g., a stock) at time t. Let D_{it} be its dividend payment between $t-1$ and t. Then, the **net** return of this asset at t is:

$$R_{it} = \frac{P_{it} + D_{it} - P_{it-1}}{P_{t-1}} = \frac{P_{it} + D_{it}}{P_{it-1}} - 1.$$

- NOTE: I use R to denote the net return of an asset, while in some (and maybe many) other materials, R is used to denote the **gross** return, which is $1 +$ net return.

- Let R_f denote the risk-free rate and assume it is constant.

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Log Return

- Let $p_{it} = \log P_{it}$. Then, the **logarithmic** return or continuously compounded return on an asset is:

$$r_{it} = \log(P_{it} + D_{it}) - \log(P_{it-1}).$$

- Relationship between R_{it} and r_{it}:
  - $r_{it} = \log(1 + R_{it})$, or
  - $R_{it} = \exp(r_{it}) - 1$.

  Hint: Make sure you are familiar with this transformation.

- When your time inverval from $t-1$ to t is small (e.g., you are calculating returns at daily or higher frequencies), then R_{it} and r_{it} are very close to each other.

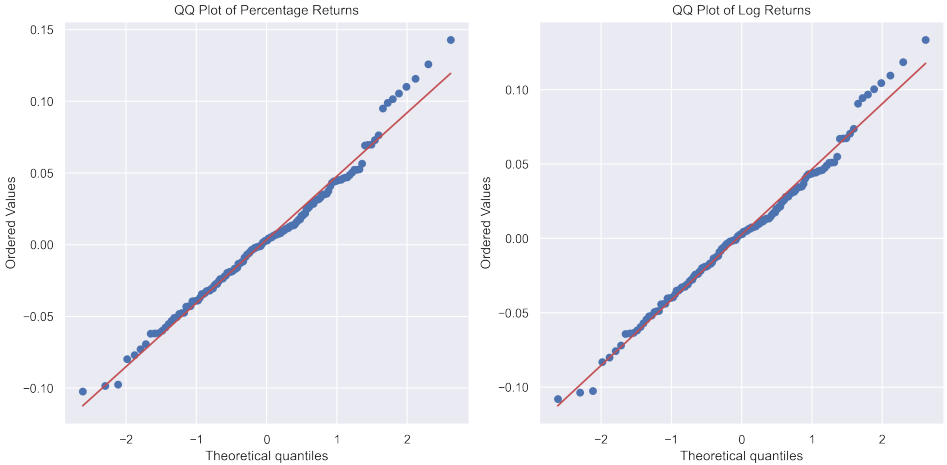
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- GOOG, 2020/11/30 to 2023/11/30, daily return



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- GOOG, 2020/11/30 to 2023/11/30, weekly return



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- If you borrow a dollar at the risk-free rate R_f and invest it in an asset with return R_i, you get the payoff $R_{it} - R_f$. This is called **excess** return.
- In asset pricing, factors are constructed as long/short portfoilos of assets (i.e., stocks). This long/short portfolio also generates an excess return. We often use the notation R^e_{it} for excess returns.
- Asset pricing is about how we can find a model for the expected excess return, i.e., $\mathbb{E}[R^e_i]$.

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- Notation:
  - $\boldsymbol{R}$ ($N \times 1$ vector): asset return.
  - $\boldsymbol{\mu}$ ($N \times 1$ vector): expected returns of the assets, i.e., $\boldsymbol{\mu} = \mathbb{E}[\boldsymbol{R}]$.
  - $\boldsymbol{\Sigma}$ ($N \times N$ matrix): variance-covariance matrix of $\boldsymbol{R}$.
  - $\mathbf{w}$ ($N \times 1$ vector): weights of assets in the portfolio.
  - Portfolio's expected return: $\boldsymbol{w}^\top \boldsymbol{\mu}$.
  - Portfolio's variance: $\boldsymbol{w}^\top \boldsymbol{\Sigma} \boldsymbol{w}$.

- The mean-variance analysis can be stated as "***choose a portfolio to minimize variance for a given mean***":

$$\begin{aligned} \min_{\boldsymbol{w}} \quad & \boldsymbol{w}^\top \boldsymbol{\Sigma} \boldsymbol{w} \\ \text{subject to} \quad & \boldsymbol{w}^\top \boldsymbol{\mu} = \mu^\star \\ & \boldsymbol{w}^\top \mathbf{1} = 1. \end{aligned}$$

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- The optimal weights:

$$\boldsymbol{w}^\star = \frac{C\mu^\star - B}{AC - B^2}\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu} + \frac{A - B\mu^\star}{AC - B^2}\boldsymbol{\Sigma}^{-1}\mathbf{1}.$$

where $A = \boldsymbol{\mu}^\top\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}$, $B = \boldsymbol{\mu}^\top\boldsymbol{\Sigma}^{-1}\mathbf{1}$ and $C = \mathbf{1}^\top\boldsymbol{\Sigma}^{-1}\mathbf{1}$.

LET'S DO THE MATH.

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- Re-write the expression as:

$$\begin{aligned} \boldsymbol{w}^\star &= \frac{C\mu^\star - B}{AC - B^2} B \frac{\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}}{B} + \frac{A - B\mu^\star}{AC - B^2} C \frac{\boldsymbol{\Sigma}^{-1}\mathbf{1}}{C} \\ &= \left(\frac{BC\mu^\star - B^2}{AC - B^2}\right) \frac{\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}}{\boldsymbol{\mu}^\top \boldsymbol{\Sigma}^{-1}\mathbf{1}} + \left(\frac{AC - BC\mu^\star}{AC - B^2}\right) \frac{\boldsymbol{\Sigma}^{-1}\mathbf{1}}{\mathbf{1}^\top \boldsymbol{\Sigma}^{-1}\mathbf{1}}. \end{aligned}$$

- $\boldsymbol{w}^\star$ is the weighted average of two special portfolios $\boldsymbol{w}_1 = \frac{\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}}{\boldsymbol{\mu}^\top \boldsymbol{\Sigma}^{-1}\mathbf{1}}$ and $\boldsymbol{w}_2 = \frac{\boldsymbol{\Sigma}^{-1}\mathbf{1}}{\mathbf{1}^\top \boldsymbol{\Sigma}^{-1}\mathbf{1}}$, and it is uniquely determined by $\mu^\star$.

- This phenomenon is also known as the **two-fund separation theorem**.

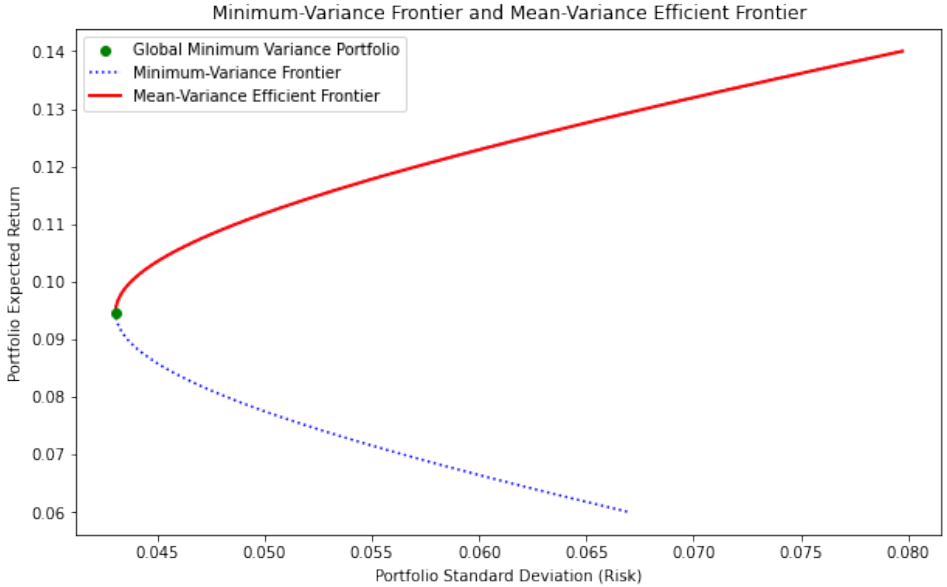
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- Regardless of individual risk preferences (i.e., different choices of $\mu^\star$), all investors will hold a combination of the same two basic portfolios.

- The basis portfolios are not unique, but $\boldsymbol{w}_1$ and $\boldsymbol{w}_2$ mentioned above have some special properties.
  - $\boldsymbol{w}_1$ gives you the highest risk-adjusted expected return that you can achieve by investing in all risky assets.
  - $\boldsymbol{w}_2$ is the global minimum variance portfolio.

- Solve for different $\boldsymbol{w}^\star$ for different $\mu^\star$, and calculate its standard deviation $\sigma_p = \sqrt{(\boldsymbol{w}^\star)^\top \boldsymbol{\Sigma} \boldsymbol{w}^\star}$. The pairs (σ_p, μ_p) form the **minimum-variance frontier** of the **risky** assets, with the upper half of this hyperbolic curve being the **mean-variance efficient frontier** in the absense of a risk-free asset.

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- Suppose there is a risk-free asset R_f.
- Let $\boldsymbol{w}$ represent the weights of risky assets, and then the weight on the risk-free asset is $1 - \boldsymbol{w}^\top \mathbf{1}$:
  1. Weights of all assets including both risky and risk-free assets sum to 1.
  2. $1 - \boldsymbol{w}^\top \mathbf{1}$ can be negative, i.e., you borrow money at the risk-free rate to invest in the risky assets.
- The expected value of this portfolio is $\boldsymbol{w}^\top \boldsymbol{\mu} + (1 - \boldsymbol{w}^\top \mathbf{1}) R_f$ and the objective function becomes:

$$\begin{aligned} \min_{\boldsymbol{w}} \quad & \boldsymbol{w}^\top \boldsymbol{\Sigma} \boldsymbol{w} \\ \text{subject to} \quad & \boldsymbol{w}^\top \boldsymbol{\mu} + (1 - \boldsymbol{w}^\top \mathbf{1}) R_f = \mu^\star. \end{aligned}$$

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- Rearrange the constraint and use $\boldsymbol{\mu}^e = \boldsymbol{\mu} - R_f \mathbf{1}$:

$$\begin{aligned} \min_{\boldsymbol{w}} \quad & \boldsymbol{w}^\top \boldsymbol{\Sigma} \boldsymbol{w} \\ \text{subject to} \quad & \boldsymbol{w}^\top \boldsymbol{\mu}^e = \mu^\star - R_f. \end{aligned}$$

- We no longer need the constraint $\boldsymbol{w}^\top \mathbf{1} = 1$. (Why?)

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- The optimal weights:

$$\boldsymbol{w}^\star = \left(\frac{\mu^\star - R_f}{(\boldsymbol{\mu}^e)^\top \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}^e} \right) \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}^e.$$

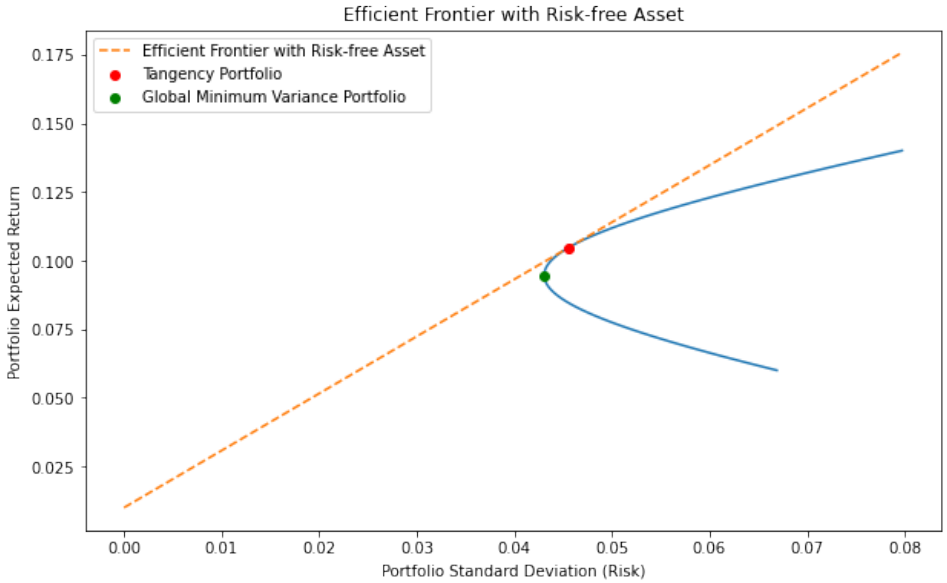
- Express $\boldsymbol{w}^\star$ as a scaled version of a special portfolio:

$$\boldsymbol{w}^\star = \underbrace{\left(\frac{(\mu^\star - R_f) \mathbf{1}^\top \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}^e}{(\boldsymbol{\mu}^e)^\top \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}^e} \right)}_{\delta} \underbrace{\frac{\boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}^e}{\mathbf{1}^\top \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}^e}}_{\boldsymbol{w}_{\tan}}.$$

- The sum of the weights in $\boldsymbol{w}^\star$, i.e., $(\boldsymbol{w}^\star)^\top \mathbf{1}$, equals δ. **We allocate δ on $\boldsymbol{w}_{\tan}$ and $1 - \delta$ on R_f.**

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- In finance, for an asset (either an individual security or a portfolio) with an expected return of μ and variance σ^2, the **Sharpe ratio** is defined as

$$\text{Sharpe Ratio} = \frac{\mu - R_f}{\sigma}.$$

- The **slope** of the tangent line indeed represents the **highest** Sharpe ratio that can be achieved by these risky assets and the risk-free asset.
- This straight line (i.e., efficient frontier) is also called the capital market line (CML). Let M denote the tangency point, then CML is defined by the following linear equation:

$$\mu = R_f + \frac{\mu_M - R_f}{\sigma_M}\sigma.$$

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- A very important result in asset pricing theory: **For any portfolio on the mean-variance frontier (other than the risk-free asset, and not necessarily the tangency portfolio), we can construct a single factor model.**

$$\mu_i - R_f = \beta_i(\mathbb{E}[R_{mv}] - R_f),$$

  where R_{mv} represents the return of some portfolio on the mean-variance frontier.

- We can derive this result from different approaches (see Cochrane 2005), but I pick one specific approach for its elegance and simplicity (I'd like to give full credit to Prof. Wang, Xi of Peking University, School of Economics, for this derivation).

LET'S DO THE MATH.

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- CAPM has the same expression as the above equation, except that the mean-variance portfolio is replaced by the market portfolio:

$$\mathbb{E}[R_i] - R_f = \beta_i(\mathbb{E}[R_m] - R_f).$$

- $\mathbb{E}[R_m] - R_f$ is therefore the risk premium of the market portfolio, known as the market factor. In this case, β_i is a measure of an asset's sensitivity to market movements. It represents the **systematic risk** of an asset relative to the overall market.

CAPM is a statement that the market portfolio is on the mean-variance frontier.

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- The Stochastic Discount Factor (SDF) is a random variable that adjusts future cash flows to their present value, reflecting the time value of money and risk preferences of investors.

- In a no-arbitrage framework, the present value of an asset can be expressed as:

$$P_t = \mathbb{E}[m_{t+1} \cdot X_{t+1}],$$

  where P_t is the price of the asset at time t, m_{t+1} is the stochastic discount factor between t and $t+1$, and X_{t+1} is the future payoff of the asset at time $t+1$.

- SDF captures the idea that investors require compensation for the time value of money and for bearing risk. It adjusts future payoffs to present values in such a way that all assets, when priced appropriately, offer no arbitrage opportunities.

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- If we invest one dollar today in the risk-free asset, we will get $1 + R_f$ at the end of next period. This means that

$$1 = \mathbb{E}[m_{t+1}(1 + R_f)],$$

or

$$\mathbb{E}[m_{t+1}] = \frac{1}{1 + R_f}.$$

- For the excess return R^e_{it}, since it is zero-payment today, its price must be zero:

$$0 = \mathbb{E}[m_{t+1} R^e_{it}].$$

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We have talked mean-variance frontier; and we have talked about SDF. Asset pricing theory tells us that they are the same thing (Cochrane 2005):

There is a discount factor of the form $m = a + bR_{mv}$ if and only if R_{mv} is on the mean-variance frontier, and R_{mv} is not the risk-free rate.

LET'S DERIVE THE RESULT.

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- The multi-factor model is a natural extension to it by considering multiple factors, instead of a single factor:

$$\mathbb{E}[R_i^e] = \boldsymbol{\beta}_i^\top \boldsymbol{\lambda},$$

where $K \times 1$ vector $\boldsymbol{\beta}_i$ represents factor exposures of asset i, and $K \times 1$ vector $\boldsymbol{\lambda}_i$ represents the risk premia of these factors.

A multi-factor model is a statement that a combination of the factors is on the mean-variance frontier.

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- In APT, the data generating process of assets' excess returns is governed by

$$\mathbf{R}_t^e = \mathbb{E}[\boldsymbol{R}^e] + \boldsymbol{\beta}\boldsymbol{v}_t + \boldsymbol{e}_t,$$

where:

- $\boldsymbol{R}_t^e$ is an $N \times 1$ vector of assets' excess returns,
- $\boldsymbol{\beta}$ is an $N \times K$ matrix of factor exposures,
- $\boldsymbol{v}_t$ is a $K \times 1$ vector of zero-mean factor innovations defined as $\boldsymbol{v}_t = \boldsymbol{f}_t - \mathbb{E}[\boldsymbol{f}_t]$ with $\boldsymbol{f}_t$ being factor realization,
- and $\boldsymbol{e}_t$ is an $N \times 1$ vector of zero-mean idiosyncratic errors, such that $\mathbb{E}[\boldsymbol{e}_t|\boldsymbol{v}_t] = 0$.

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- When factors are tradable portfolios, it can be shown that factors' risk premia $\boldsymbol{\lambda} = \mathbb{E}[\boldsymbol{f}_t]$ and

$$\mathbb{E}[R_i^e] = \boldsymbol{\beta}_i^\top \boldsymbol{\lambda}.$$

- In practice, a term α_i (representing **pricing error**) is added to match the empirical data:

$$\mathbb{E}[R_i^e] = \alpha_i + \boldsymbol{\beta}_i^\top \boldsymbol{\lambda}.$$

LET'S DO THE MATH.

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An expected return beta model is equivalent to a discount factor that is a linear function of the factors in the beta model. This is an important and central result. It gives the connection between the discount factor formulation and the expected return-beta factor model formulation common in empirical work.

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- Plugging $\mathbb{E}[R_i^e] = \alpha_i + \boldsymbol{\beta}_i^\top \boldsymbol{\lambda}$ into the data generating process and using the fact that $\mathbf{f}_t = \boldsymbol{\lambda} + \boldsymbol{v}_t$, we can re-write the DGP of assets' excess returns as:

$$\boldsymbol{R}_t^e = \boldsymbol{\alpha} + \boldsymbol{\beta} \boldsymbol{f}_t + \boldsymbol{e}_t.$$

- For individual asset i:

$$R_{it}^e = \alpha_i + \sum_{k=1}^{K} \beta_{ik} f_{kt} + e_{it},$$

where β_{ik} is the factor loading of asset i on factor k and f_{kt} is the factor k's excess return realization at time t.

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Two Approaches

- While theory provides a framework for studying assets' returns under the multi-factor model framework, it does not offer much guidance on which factors should be included on the right-hand-side of the equation.

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- While theory provides a framework for studying assets' returns under the multi-factor model framework, it does not offer much guidance on which factors should be included on the right-hand-side of the equation.
- **The first approach assumes either factors or factor exposures are known and observable**, and academia and industry take different stances in this regard.
  - Academia considers that factors are tradable portfolios that are constructed by using some firm characteristics, such as the HML and SMB factors of Fama and French (1993).
  - Industry assumes that factor loadings are observable, and uses firm characteristics as factor exposures, and factors' risk premia can be obtained by regressing assets' return on factor exposures.

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Two Approaches

- While theory provides a framework for studying assets' returns under the multi-factor model framework, it does not offer much guidance on which factors should be included on the right-hand-side of the equation.
- **The first approach assumes either factors or factor exposures are known and observable**, and academia and industry take different stances in this regard.
  - Academia considers that factors are tradable portfolios that are constructed by using some firm characteristics, such as the HML and SMB factors of Fama and French (1993).
  - Industry assumes that factor loadings are observable, and uses firm characteristics as factor exposures, and factors' risk premia can be obtained by regressing assets' return on factor exposures.
- **The second approach assumes both factors and factor exposures are latent**. In this case, statistical techniques such as PCA is employed to identify the factor model, and the factors are called statistical factors.

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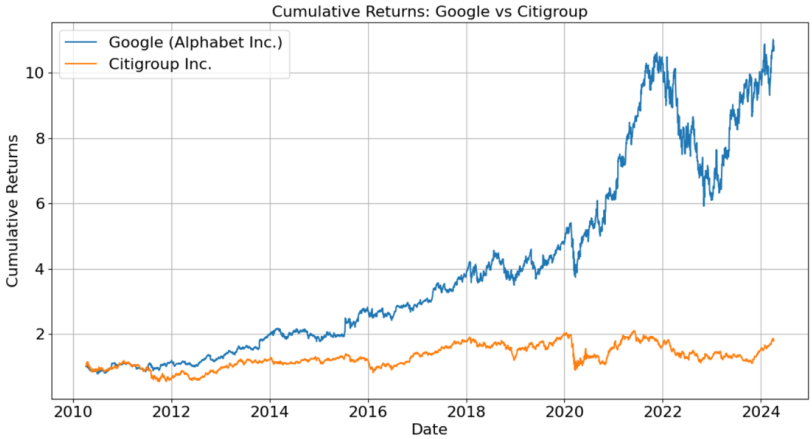
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Why do the average returns of some assets outperform the average returns of other assets?



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The Question, Cont'd

- Taking an **ex-ante** view, the expected returns of assets are different in the **cross-section**.
- Cross-section research studies what determines assets' expected returns through the lens of **factor models**.
- The development of factor models (and factor investing) is largely driven by **empirical challenges**.
- In academic language, research about the cross-section is referred to as **asset pricing**, while in industrial practice, applying factor models to earn excess returns is known as **factor investing**.

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In the 1960s, the Capital Asset Pricing Model (CAPM) was introduced. According to CAPM, the expected *excess* return of an asset is determined by the following simple linear model:

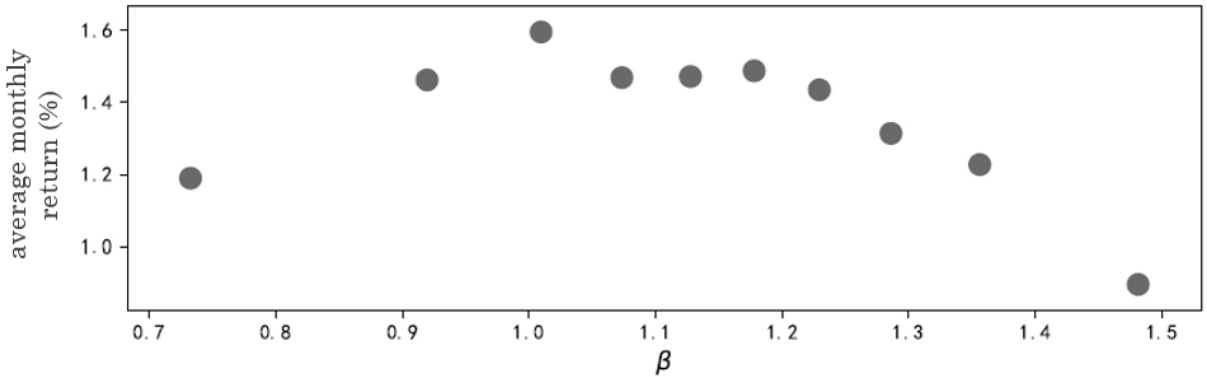
$$\mathbb{E}[R_i] - R_f = \beta_i(\mathbb{E}[R_M] - R_f),$$

where:

- $\mathbb{E}[R_i]$ and $\mathbb{E}[R_M]$ are the expected returns of asset i and market portfolio.
- R_f is the risk-free rate.
- $\beta_i = \text{cov}(R_i, R_M)/\text{var}(R_M)$ captures the sensitivity of the asset's return to the market return, also known as the asset's **exposure** to market risk.
- $\mathbb{E}[R_M] - R_f$ is referred to as the **risk premium** of the market factor.

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Expected excess returns do **NOT** align with market betas.



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- Researchers discovered that the expected excess returns of different assets are not determined solely by a single market factor but are also influenced by other factors.

- Ross (1976) proposed the well-known **Arbitrage Pricing Theory** (APT), which extended the CAPM framework by developing linear factor models. The model assumes that the expected excess return of asset i is determined by the following multivariate linear equation:

$$\mathbb{E}[R_i^e] = \boldsymbol{\beta}_i^\top \boldsymbol{\lambda},$$

  where $\mathbb{E}[R_i^e]$ represents the expected excess return of asset i, $\boldsymbol{\beta}_i$ is the factor exposure (or factor loading) of asset i, and λ represents the expected excess returns of factors, also referred to as factor risk premiums.

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It's Mainly about the Cross-Section

- Similar to CAPM, the multi-factor model assumes that the expected excess return $\mathbb{E}[R_i^e]$ of an asset is determined by the risk premiums of a series of factors, as well as the asset's exposures to these factors.
- **A multi-factor model examines the differences in expected returns across different assets rather than the time-series variations in a single asset's returns.**
- These differences in expected returns across assets are typically referred to as **cross-sectional** differences, and the expected return of an asset is determined by the magnitude of its exposure $\boldsymbol{\beta}_i$ to these factors.

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The Pricing Error α

The factor model is mathematically simple and elegant, yet real financial markets are highly complex. In practice, the left and right sides of equation often do not match, leading to a pricing error term α_i:

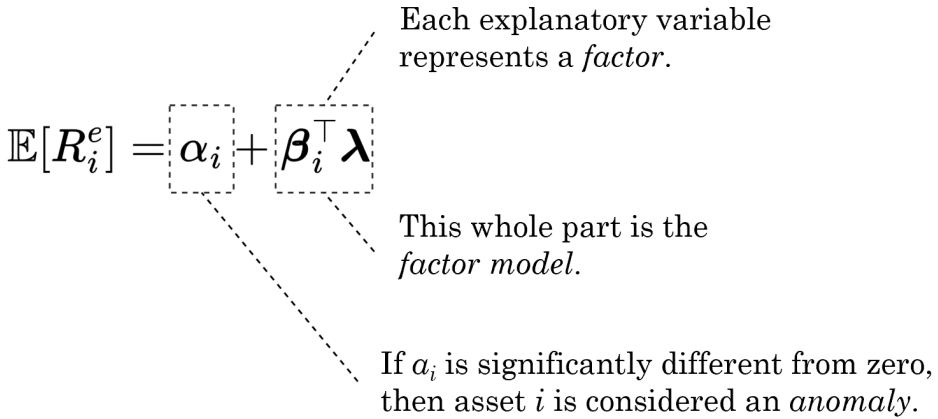
$$\mathbb{E}[R_i^e] = \alpha_i + \boldsymbol{\beta}_i^\top \boldsymbol{\lambda}.$$

where α_i represents the pricing error, also known as the **abnormal return**. Two potential reasons:

1. **Model Specification Bias**: This occurs when important factors are omitted. Adding these missing factors eliminates the error.
2. **Sampling Error**: The model itself may be correctly specified, but since the actual return data are merely a sample of the population, errors are inevitable due to simpling uncertainty.
   - If α_i does not significantly deviate from zero, it can be assumed that its occurrence is due to noise.
   - If α_i significantly deviates from zero, it indicates that the asset is mispriced.

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Put Everything Together



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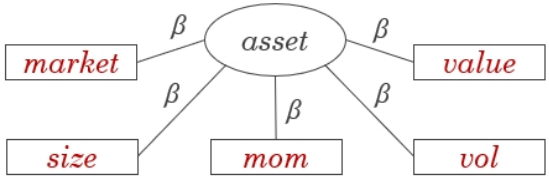
A formal definition:

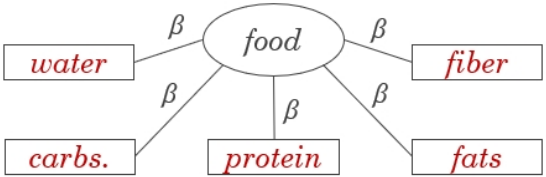
> **A factor describes a type of systematic risk to which many assets are commonly exposed; this risk is the driving force behind asset returns. The factor return is the risk premium or risk compensation for this systematic risk, representing the common return of these assets.**

Still lack of intuition? Let's look at an analogy.

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- **Factors are to assets** as **nutrients are to food**.

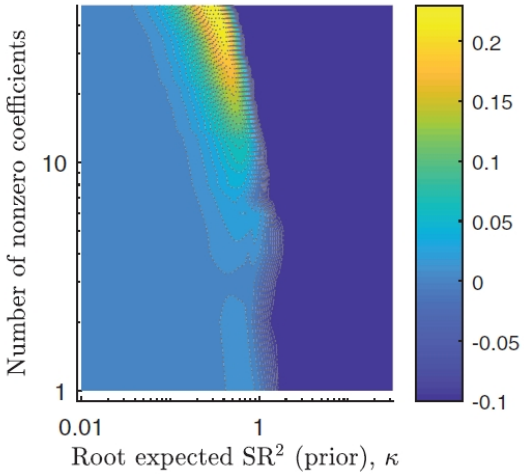




- A food is a combination of different nutrients; An asset is a combination of different factors.
- Two necessary conditions for a factor:
  1. **A factor drives the common co-movement of asset returns**, meaning that the factor must be related to the covariance matrix of asset returns;
  2. **A factor must be priced.**

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- Factors form the factor model, corresponding to the $\boldsymbol{\beta}_i^\top \boldsymbol{\lambda}$ component in the core equation.
- Traditional research follows the Law of Parsimony, i.e., preference for simplicity. Examples: ALL **ad-hoc** factor models, such as Fama-French three-factor model.
- **However, are factor models sparse?**



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- When the pricing error α_i of an asset statistically differs from zero, the asset is an anomaly.
  - For example, you may form a long/short hedged portfolio by using some sorting variable. If this portfolio has a significantly positive α_i, then the portfolio is termed an anomaly, and the variable used to construct the portfolio is called an anomaly variable.

- Anomalies (if they are real rather than results of data mining) largely challenge **market efficiency**.

- Fama: This is a **joint hypothesis problem**, i.e., to test market efficiency, you need to simultaneously specify the pricing model.
  - Side effect: Empirical research (in the past) seeks to find the factor model that **minimizes** pricing errors **in-sample**.

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A Side Note: Terminology Confusion

- Academia tends to rigorously distinguish between anomalies and factors. Following the law of parsimony, academia typically selects a limited number of factors, while viewing the rest as anomalies.
- In industry, the use of terminology is loose. There is often less emphasis on distinguishing between anomalies and factors, with both being broadly referred to as "factors". The industry is primarily concerned with whether a factor can generate excess returns against a chosen benchmark (e.g., the S&P500 Index).
- Understanding the difference in how academia and the industry view factors and anomalies sets the stage for exploring the key elements of factor investing.

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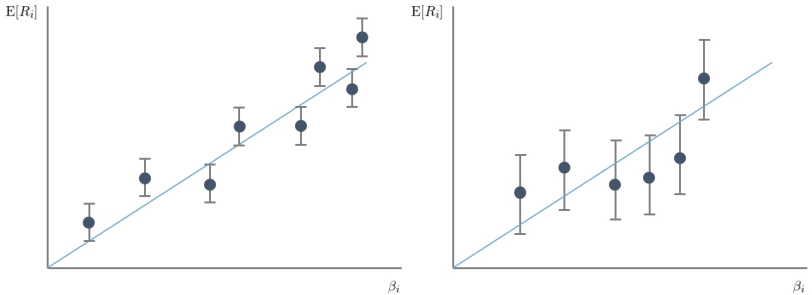
Research about $\beta_i^\top \lambda$

- Academia:
  - Identify the "best" factor model, e.g., Fama-French models, Hou-Xue-Zhang q-factor model, etc.
  - Develop econometric methods to estimate factor exposures, factor risk primiums, and conduct statistical testing on factor models.
  - Performance attribution for active fund managers.

- Industry:
  - The primary goal is to use factors to achieve returns that exceed a benchmark.
  - From an *asset allocation* perspective, industry focuses on identifying factors with positive risk premiums over the long term and allocating as much exposure (i.e., high β_i) as possible to these factors, under the premise of **diversification**.
  - Numerous Smart Beta ETFs have been introduced, greatly enriching the tools available for asset allocation.

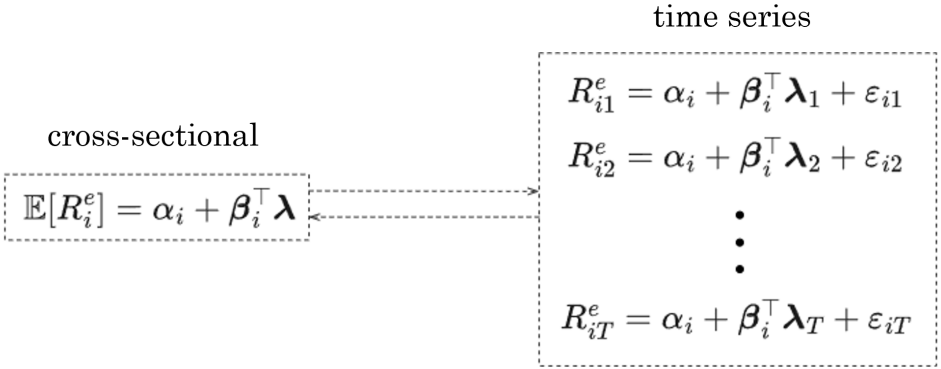
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- Academia:
  - Anomalies can be used to test whether the market is efficient. This is the primary motivation behind academic research on anomalies.
  - Anomalies are used to compare different factor models. For example, if there are fewer anomalies under Model than under Model B, then Model A is considered better.
  - **BUT...**



- Industry: Utilize anomalies to earn excess returns after transaction costs.

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- In matrix form:

$$\mathbf{R}_t^e = \boldsymbol{\alpha} + \boldsymbol{\beta}\boldsymbol{\lambda}_t + \boldsymbol{\varepsilon}_t,$$

where $\mathbf{R}_t^e = [R_{1t}^e, R_{2t}^e, \cdots, R_{Nt}^e]^\top$; $\boldsymbol{\alpha} = [\alpha_1, \alpha_2, \cdots, \alpha_N]^\top$; $\boldsymbol{\beta} = [\boldsymbol{\beta}_1, \boldsymbol{\beta}_2, \cdots, \boldsymbol{\beta}_N]^\top$; and $\boldsymbol{\varepsilon}_t = [\varepsilon_{1t}, \varepsilon_{2t}, \cdots, \varepsilon_{Nt}]^\top$ satisfying $\mathbb{E}[\boldsymbol{\varepsilon}_t] = \mathbf{0}$ and $\text{cov}(\boldsymbol{\lambda}_t, \boldsymbol{\varepsilon}_t) = \mathbf{0}$.

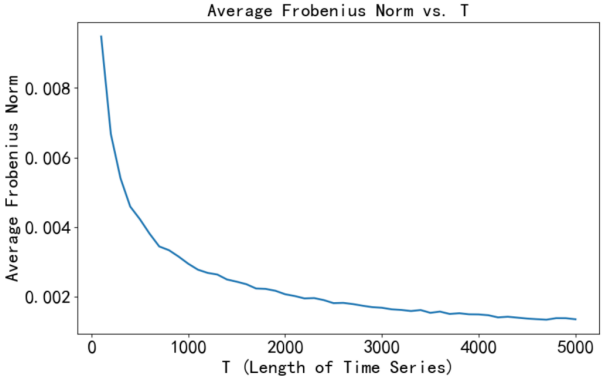
- Taking the covariance matrix of both sides and using $\text{cov}(\boldsymbol{\lambda}_t, \boldsymbol{\varepsilon}_t) = \mathbf{0}$:

$$\boldsymbol{\Sigma} = \boldsymbol{\beta}\boldsymbol{\Sigma}_\lambda\boldsymbol{\beta}^\top + \boldsymbol{\Sigma}_\varepsilon,$$

where $N \times N$ matrix $\boldsymbol{\Sigma}$ is the covariance matrix of those N assets, $K \times K$ matrix $\boldsymbol{\Sigma}_\lambda$ is the covariance matrix of those K factors, and $N \times N$ matrix $\boldsymbol{\Sigma}_\varepsilon$ is the covariance matrix of the error terms. Since ε_{it} are mutually independent, $\boldsymbol{\Sigma}_\varepsilon$ is a diagonal matrix.

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- Academia:
  - $\boldsymbol{\Sigma} = \boldsymbol{\beta}\boldsymbol{\Sigma}_\lambda\boldsymbol{\beta}^\top + \boldsymbol{\Sigma}_\varepsilon$ implies that factors should be related to the covariance matrix of assets.
  - *Statistical factors* are derived from this covariance matrix relationship by means of PCA in recent years.
- Industry:
  - Risk Management: estimation of the covariance matrix. Without a factor model, we require $T \geq N$ as otherwise the sample covariance matrix is non-invertible.
  - Factor Timing: factor returns fluctuate over time; *conditional* risk premiums vary.



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Academic research can be broadly categorized into theoretical and empirical aspects:

- Theoretical:
  - Statistical methods were developed to test anomalies and factors, and to compare factor models (e.g., Fama-MacBeth regression, GRS test, etc.).
  - With the rapid development of machine learning, more scholars are applying these techniques to the development of factor models.
- Empirical:
  - Academic effort to identify new factors has been aggressive.
  - Some findings enriched our understanding of the true drivers behind stock returns.
  - Some discoveries are merely products of in-sample overfitting, offering little benefit to factor investing.

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Stage 1: CAPM (1960s)

- The **Capital Asset Pricing Model (CAPM)**:

$$\mathbb{E}[R_i] - R_f = \beta_i(\mathbb{E}[R_M] - R_f).$$

- Eugene Fama: *The arrival of the CAPM was like the time after a thunderstorm, when the air suddenly clears.*

- However, some assets earn abnormal returns against CAPM:
  - Value stocks outperform growth stocks (Basu 1977)
  - Small-cap stocks outperform large-cap stocks (Banz 1981)

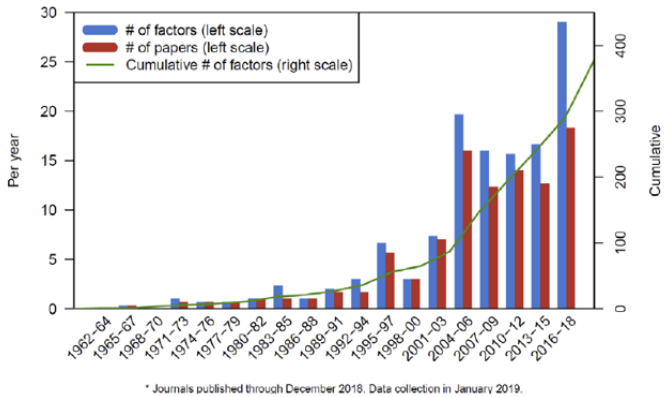
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- Fama and French (1993) three-factor model:

$$\mathbb{E}[R_i^e] = \beta_{i,M}(\mathbb{E}[R_M] - R_f) + \beta_{i,\text{HML}}\mathbb{E}[R_{\text{HML}}] + \beta_{i,\text{SMB}}\mathbb{E}[R_{\text{SMB}}].$$

- *Zoo of factors* (Cochrane 2011):
  1. Which characteristics provide independent information?
  2. Does each new anomaly variable also correspond to a new factor?
  3. How many of these new factors are really important?

Figure 1. Out of control factor production - through January 2019



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- Many **ad-hoc parsimony** factor models are proposed since 2010.

Model	Factors
US Stock Market	
Fama and French (1993)	Market, Size, Value
Carhart (1997)	Market, Size, Value, Momentum
Novy-Marx (2013)	Market, Value, Momentum, Profitability
Fama and French (2015)	Market, Size, Value, Profitability, Investment
Hou, Xue and Zhang (2015)	Market, Size, Profitability, Investment
Stambaugh and Yuan (2017)	Market, Size, Management, Performance
Daniel, Hirshleifer and Sun (2020)	Market, Long-Term behavioral, Short-Term behavioral
Chinese Stock Market	
Liu, Stambaugh, and Yuan (2019)	Market, Size, Value

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- Empirical Challenge 1: **The high dimensionality of covariates.**

Data Type	Representative Papers
Trading account	Barber and Odean (2000), Boehmer et al. (2008), Kaniel et al. (2008), Kelley and Tetlock (2013), Barrot et al. (2016)
Order book	Goettler et al. (2009), Cont et al. (2014)
Social media	Antweiler and Frank (2004), Tetlock (2007), Chen et al. (2014), Jame et al. (2016), Bartov et al. (2018), Leung et al. (2019), Umar (2022)
Crowdsourced data	Green et al. (2019), Da et al. (2021)
Text data	Cohen et al. (2020), Bekkerman et al. (2023), Garcia et al. (2023), Bybee et al. (forthcoming)
Image data	Zhu (2019), Obaid and Pukthuanthong (2022), Jiang et al. (2023)
Weather data	Hirshleifer and Shumway (2003), Goetzmann et al. (2014)
Web search data	Da et al. (2011), Drake et al. (2012), Wu and Brynjolfsson (2015)

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- Empirical Challenge 2: **Nonlinear relationship between covariates and asset returns.**

	Highest IVOL	Next 20%	Next 20%	Next 20%	Lowest IVOL	Highest –Lowest	All Stocks
Most Overpriced	−1.89	−0.95	−0.72	−0.47	−0.39	−1.50	−0.81
(Top 20%)	(−12.05)	(−7.39)	(−4.90)	(−3.62)	(−3.04)	(−7.36)	(−8.14)
Next 20%	−0.88	−0.41	−0.31	−0.21	−0.04	−0.84	−0.23
	(−5.86)	(−3.36)	(−3.00)	(−2.08)	(−0.44)	(−4.41)	(−3.88)
Next 20%	−0.09	−0.01	−0.05	−0.12	0.02	−0.10	−0.07
	(−0.53)	(−0.09)	(−0.48)	(−1.29)	(0.18)	(−0.53)	(−1.47)
Next 20%	−0.15	0.07	0.17	0.18	0.23	−0.38	0.18
	(−0.80)	(0.63)	(1.87)	(2.33)	(3.22)	(−1.78)	(4.45)
Most Underpriced	0.56	0.68	0.51	0.33	0.14	0.41	0.28
(Bottom 20%)	(3.27)	(4.91)	(5.02)	(4.10)	(2.04)	(2.16)	(5.67)
Most Overpriced –	−.44	−1.63	−1.23	−0.81	−0.53	−1.91	−1.09
Most Underpriced	(−11.07)	(−8.65)	(−6.43)	(−5.02)	(−3.43)	(−7.62)	(−8.05)
All Stocks	−0.69	−0.12	−0.00	0.05	0.08	−0.78	
	(−6.09)	(−1.56)	(−0.01)	(1.07)	(1.86)	(−5.50)	

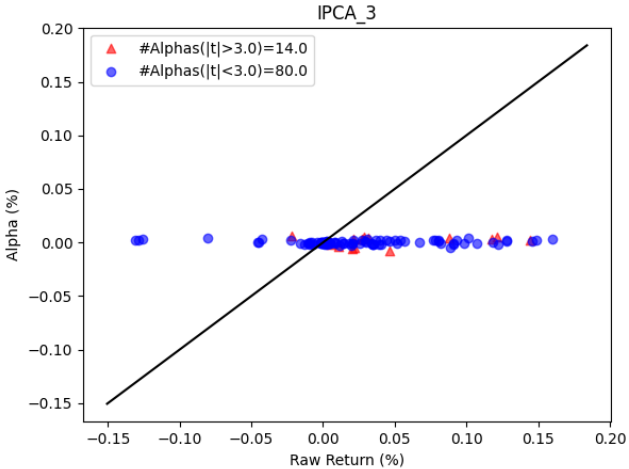
Source: Stambaugh, Yu and Yuan (2015)

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- The empirical challenges shift research norm from **econometrics** to **machine learning**.
- Latent factor model, i.e., modeling $\boldsymbol{\beta}$:

$$R^e_{i,t+1} = \boldsymbol{\beta}^\top_{i,t}(\mathbf{c})\boldsymbol{f}_{t+1} + e_{i,t+1}.$$

  - factors are latent in this framework
  - machine learning view: $\boldsymbol{\beta}(.)$ maps a **large** information set to a small factor model



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- 1950s, chaos
- 1960s, CAPM $\mathrm{E}[R_i] - R_f = \beta_i(\mathrm{E}[R_M] - R_f)$
- 1980s, anomalies of CAPM (e.g., size, value), chaos again

  $$\mathrm{E}[R_i] - R_f = \alpha_i + \beta_i(\mathrm{E}[R_M] - R_f)$$
- 1990s, Fama-French three factor model

  $$\mathrm{E}[R_i^e] = \beta_i(\mathrm{E}[R_M] - R_f) + h_i\mathrm{E}[R_{HML}] + s_i\mathrm{E}[R_{SMB}]$$
- 2000s, zoo of factors, chaos again

  $$\mathrm{E}[R_i^e] = \alpha_i + \beta_i(\mathrm{E}[R_M] - R_f) + h_i\mathrm{E}[R_{HML}] + s_i\mathrm{E}[R_{SMB}]$$
- 2010s, "factor war"(e.g., FF5, q-factor model, etc)

  $$\mathrm{E}[R_i^e] = \beta_i'\lambda$$
- 2020s, a new research norm in the era of machine learning

  $$R_{i,t+1}^e = \beta_{i,t}'F_{t+1} + e_{i,t+1}$$

  $\beta()$: linear/nonlinear func of firm characteristics; map a large info set to a small factor model

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Questions?

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